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United States Patent	12388793
Kind Code	B2
Date of Patent	August 12, 2025
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Supporting local host traffic in remote device infrastructure

Abstract

A remote software development infrastructure can include multiple cross-continent datacenters with a variety of remote devices to choose from in the datacenters. A user can select a remote device and access the remote device via a local machine. The infrastructure can stream a video feed of the display of the remote device to the local machine. The user can run a browser on the remote device to test a website, including for example, testing accessing the website and its features via the browser running on the remote device. URL requests to the user's website are received from the local machine and injected into the browser running on the remote device. The browser running on the remote device issues the URL requests. Restricted URL requests can be modified via a browser extension and replaced with a non-restricted URL pointing to the same communication endpoint as the restricted URL.

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Appl. No.: 18/373796

Filed: September 27, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240205193 A1	Jun. 20, 2024

Related U.S. Application Data

continuation parent-doc US 18082772 20221216 US 11888821 child-doc US 18373796

Publication Classification

Int. Cl.: H04L29/08 (20060101); G06F9/54 (20060101); G06F16/955 (20190101); H04L9/40 (20220101)

U.S. Cl.:

CPC H04L63/0281 (20130101); G06F16/9566 (20190101); H04L63/0236 (20130101); H04L63/0272 (20130101);

Field of Classification Search

CPC: H04L (63/0281); H04L (63/0236); H04L (63/0272); G06F (16/9566)

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Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/082,772, filed on Dec. 16, 2022, titled “SUPPORTING LOCAL HOST TRAFFIC IN REMOTE DEVICE INFRASTRUCTURE,” which is hereby incorporated in its entirety and should be considered a part of this disclosure.

BACKGROUND

Field

(1) This invention relates generally to the field of enabling a remote infrastructure for application and website development on multiple platforms, and more particularly to testing a local network using the remote infrastructure.

Description of the Related Art

(2) The approaches described in this section are approaches that could be pursued, but not necessarily approaches that have been previously conceived or pursued. Therefore, unless otherwise indicated, it should not be assumed that any of the approaches described in this section qualify as prior art merely by virtue of their inclusion in this section.

(3) The multitude of computers, mobile devices and platforms have given consumers a vast universe of choices. Naturally, software, application and website developers have a keen interest in ensuring their products work seamlessly across the existing platforms, including older devices on the market. This creates a challenge for the developers to properly test their products on the potential devices and platforms that their target consumer might use. On the one hand, acquiring and configuring multiple potential target devices can strain the resources of a developer. On the

other hand, the developer cannot risk disregarding a potential target device in his typical development cycle. Even for prominent platforms, such as IOS® and Android®, at any given time, there are multiple generations and iterations of these devices on the market, further complicating the development and testing process across multiple platforms. This dynamic illustrates a need for a robust infrastructure that enables developers to test their products across multiple devices and platforms, without having to purchase or configure multiple devices. Described systems and techniques enable software developers to test their devices across multiple platforms, without having to invest resources in the hardware, software and configuration needed to perform multi-platform development.

SUMMARY

(4) The appended claims may serve as a summary of this application.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These drawings and the associated description herein are provided to illustrate specific embodiments of the invention and are not intended to be limiting.

(2) FIG. 1 illustrates a diagram of an infrastructure, enabling a software developer user to perform local software development, using remote devices.

(3) FIG. 2A illustrates an option menu, via which a user developer can choose a smartphone and a corresponding mobile browser, to test software.

(4) FIG. 2B illustrates a diagram of a user local browser and a remote device browser during a test session of software.

(5) FIG. 3 illustrates a diagram of an example of operations and components of the embodiment of FIG. 1 to avoid an exclusion list of a browser running on the remote device.

(6) FIG. 4 illustrates an example method of operation of the embodiment of FIG. 1.

(7) FIG. 5 illustrates an example method of avoiding the implicit bypass rules of a browser.

(8) FIG. 6 illustrates an example method of routing traffic and URL requests in the embodiment of FIG. 1.

(9) FIG. 7 illustrates an environment in which some embodiments may operate.

DETAILED DESCRIPTION

(10) The following detailed description of certain embodiments presents various descriptions of specific embodiments of the invention. However, the invention can be embodied in a multitude of different ways as defined and covered by the claims. In this description, reference is made to the drawings where like reference numerals may indicate identical or functionally similar elements.

(11) Unless defined otherwise, all terms used herein have the same meaning as are commonly understood by one of skill in the art to which this invention belongs. All patents, patent applications and publications referred to throughout the disclosure herein are incorporated by reference in their entirety. In the event that there is a plurality of definitions for a term herein, those in this section prevail. When the terms “one”, “a” or “an” are used in the disclosure, they mean “at least one” or “one or more”, unless otherwise indicated.

(12) A critical aspect of developing software is to test the software on a variety of platforms, operating systems, and devices on which the software is going to be running. Another aspect of developing software is to develop the software in a non-public setting before launch to ensure the software meets its production goals. Local development is often used to program and debug software. In a local development environment, the development team may operate from different geographical regions, logging into a private network to code and develop the software. The private network may be closed off from the rest of the internet or may interact with the outside world through controlled mechanisms and protocols to ensure security, privacy and confidentiality of the

development environment. Naturally, software development in many cases, require both interaction with public servers on the internet, and testing the software on a multitude of platforms and devices making calls and interacting with private and public sources. For example, the developer a website may wish to test the interactions of the website with both public servers, accessible on the Internet and private servers accessible only through the internal network of the software developer.

(13) FIG. 1 illustrates a diagram of an infrastructure **100**, enabling a software developer user to perform local software development, using remote devices. The user may establish one or more instances of a private or user test network **102**, to develop software **103** among the user team members. Using the infrastructure **100**, the software **103** can be run on a remote device **112**. The remote device **112** can be located in a datacenter **116**, where multiple remote devices **112** are available for the user to select. The datacenter **116** can provide a variety of types, operating systems, and versions of the remote devices **112** for the user to select. In the cases where the software **103** is a website or a web application, the software **103** can run on a remote device browser (RDB) **109**. While the embodiments will be described in reference to the software **103** being a website or a web application, the same technology is applicable to the development of other software programs on the infrastructure **100**. The user can access the infrastructure **100** via a remote-enabled local (REL) application **106**, provided by the owner or operator of the infrastructure **100**. The REL application **106** can be a web application, running on the user local browser (ULB) **105**. Alternatively, the REL application **106** can be a desktop application.

(14) The infrastructure **100** can transmit user's input commands from the ULB **105** to the datacenter **116** and inject the commands to the remote device **112** and to the software **103** running on the remote device **112**. The process of injecting the user commands can include translating the user input commands to input format compatible with the remote device **112**. Furthermore, the user's input commands are entered on a replica display of the remote device **112**, generated on the ULB **105**, as will be described herein. The infrastructure **100** can provide a video stream of the display of the remote device **112** and display responses of the software **103** to the ULB **105**. The user can inspect the responses of the software **103** on the ULB **105**. A variety of software and hardware components of the infrastructure **100** can provide the functionality of transmitting user input commands to the software **103**, running on the remote device **112** and generating a replica or mirrored display of the remote device **112** on the ULB **105**. In some embodiments, the video stream of the display of the remote device **112** is used to generate a replica or mirrored image of the remote device **112** on the ULB **105**. The user input commands can be received in the form of interactions with the replica display on the ULB **105**. For example, when the remote device **112** is a smart phone, with a touch screen, the user can input taps and touch screen gestures, via mouse clicks, on-screen keyboard inputs, via a laptop or desktop keyboard and so forth. The infrastructure **100** can capture the user interactions with the replica display and transmit them to the remote device **112**.

(15) The remote device **112** is connected to the Internet via a host **118** via wired and/or wireless communication. For example, the host **118** can be connected to the remote device **112** via a universal serial bus (USB) connection. The host **118** can be a desktop, laptop or a hardware server compatible with the remote device **112**. For example, Windows®, Macintosh®, and Unix® machines can be used to implement the host **118**. In some embodiments, the infrastructure **100** can include a remote device application (RDA) **111** installed on the remote device **112**, which assists in enabling the operations of the infrastructure **100**. The RDA **111** can include a variety of components and processes. A communication module **122** can be used to establish a communication network between the remote device **112** and the ULB **105**. For example, in some embodiments, a peer-to-peer communication network over the Internet can be used to establish communication between the remote device **112** and a user local machine running the ULB **105**.

(16) The communication network between the ULB **105** and the remote device **112** can include various communication channels, such as video and data communication channels. The video

communication channel is used to stream a video feed of the display of the remote device **112** to the ULB **105** and to generate a replica display of the remote device **112** on the ULB **105**. The data communication channel is used to capture and transmit the user interactions inputted to the replica display to the remote device **112**. The user can interact with the replica display, as if it had active and interactive components. For example, the user can click on a text field and enter text. The user can click the buttons of an application or website mirrored on the replica display. In the case of a webpage, the user can scroll the replica display, using mouse or keyboard inputs.

(17) An interaction server **120** on the datacenter **116** side can translate the user interactions to input gestures compatible with the remote device **112** and input those translated gestures to the remote device **112**, which can in turn input the translated gestures to the software **103**. A screen capture application **124** can capture a video feed of the display of the remote device **112** and can transmit the video feed via the video communication channel to the ULB **105**. The ULB **105** or an application running on the ULB **105** can use the video feed to generate the replica display. The user interacts with the replica display, and no parallel processes of the software **103** are run on the ULB **105** or the local machine on which the ULB **105** is installed. The RDA **111** can include additional components, such as a media capture module **126**, which can assist in capturing user input media, such as voice in instances where the user may wish to inject or test media input to the software **103**.

(18) The user test network **102** can include one or more user servers **104** to handle software calls, APIs and other development needs of software **103**, locally within the user test network **102**. The user test network **102** can be a private network inaccessible to and from the outside networks, or it can be semi-private and connected with public networks, such as the Internet. An operator of the infrastructure **100** can provide a remote-enabled local (REL) application **106**, which the user can receive and run in the user test network **102**. The operator can further provide a repeater **110** and remote devices **112** to enable the user to develop the software **103**, using a remote device **112**. The operator in most cases is independent from the user developer team and may have no access to the user test network **102**, other than commands and access granted by the user through the REL application **106**. The software **103** may be a website and/or a web application, running on the RDB **109**. If the remote device **112** is a smartphone or mobile computing device, the RDB **109** can be a mobile browser, such as Chrome®, Safari®, or another brand of mobile browser. Through tunneling connections, the ULB **105** and the REL application **106** connect with the remote device **112** and the RDB **109**, where the RDB **109** issues the software **103** traffic requests on and from the remote device **112**. A video feed of the display of the remote device **112** is captured and generated on the ULB **105**, for the user to see and inspect the display of the remote device **112**. Besides the display of the remote device **112** generated on the replica display on the ULB **105**, the infrastructure **100** can also provide the user with various reports and logs and other test and development data about the connections between the ULB **105** and the remote device **112** and/or about the internal processes of the remote device **112**, as they relate to the software **103**. Test and development data may be accessible via a user interface (UI) element, such as a developer toolbar, as a feature of the REL application **106**.

(19) The infrastructure **100** does not run parallel processes at the remote device **112** and the user test network **102**, nor does the infrastructure **100** generate mirrored copies of data or processes between the two. In other words, the software **103** does not run in the user test network **102**; rather, the remote device **112** runs the software **103**. For example, the traffic requests of the software **103** are issued from the remote device **112** and the responses from a public server **114** and/or the user test network **102** are tunneled through the infrastructure **100** and relayed to the remote device **112** and ultimately to the software **103**. In other words, the same data packets in a request and/or response are routed, without generating copies of the packets at the two locations. In a typical case, the user/developer may use a desktop, having a keyboard, monitor and mouse to develop and test the software **103** on a remote device **112**.

(20) The remote device **112** in a typical case may be a smartphone or tablet device, running a

mobile operating system. The infrastructure **100** injects the interactions of the user developer, received on the ULB **105**, into the remote device **112**. The infrastructure **100** also mirrors the screen output of the remote device **112** on the ULB **105** by streaming a video feed of the display of the remote device **112** to the ULB **105**. The mirroring operations of the infrastructure **100** can be considered near real time, save for typical network delays that may be present. Still, the interactions between the user developer and mirrored display of the remote device **112** in the user developer's local machine is seamless and appears, without perceptible delay. The user developer receives a display of a selected remote device **112** on the developer's local monitor and uses keyboard, mouse, or other input devices to interact with the mirrored or replicated display of the remote device **112**, as if the remote device **112** were locally present. The user developer's inputs received via local input devices (e.g., keyboard, mouse, etc.) are translated to input gestures compatible with the remote device **112**. For example, when the remote device **112** is a smartphone, the user developer's inputs are translated to inputs such as swipes, slides, taps, pinches, or other smartphone gesture inputs. The display output of the remote device **112** is streamed on the local monitor of the user developer in near real time, such that the user developer interactions with the remote device **112** are seamless.

(21) A tunneling agent can establish a tunneling connection **108** between the REL application **106** and a repeater **110**. The repeater **110** acts as a proxy, receiving and routing traffic to and from the REL application **106** through the tunneling connection **108**. Under normal operating conditions, the tunneling connection **108** is ON during a test session. Furthermore, the infrastructure **100** allows the developer to test software calls to public servers **114** on the internet or other non-user-defined networks. In some implementations, the REL application **106** can alternatively be termed a binary/extension application.

(22) The remote devices **112** can be in a datacenter **116**. Multiple datacenters **116** in the form of datacenters or cloud infrastructure backbone can be deployed around the world. Each datacenter **116** can include a substantial number of products, devices, and platforms to enable thousands of users in various geographical locations to test their software on the remote devices **112**. For example, the datacenter **116** can include a variety of cellphone brands, cellphone operating systems, different models (new, midmarket, and old versions) of the cellphones, various browsers, and various operating systems, and a combination of devices hardware and software-wise. The datacenter **116** can include multiple copies of the same device and software combination to enable multiple users to use them simultaneously for development. In some embodiments, thousands of devices and platforms are housed in multiple datacenters across different geographical regions to provide the functionality described herein to thousands of software developers. Nearly a million daily test sessions across the infrastructure **100** are typical. In some embodiments, the datacenter **116** can be implemented and deployed in and as a cloud architecture.

(23) FIG. 2A illustrates an option menu, via which a user developer can choose a smartphone and a corresponding mobile browser, such as RDB **109** to test the software **103**. In this example, the user developer runs the REL application **106** locally on his machine via the ULB **105**. Alternatively, the REL application **106** may be a desktop application, not needing a browser, such as the ULB **105** to run. The infrastructure **100** can select and configure a remote device **112** and a browser RDB **109** according to the user's selection. In FIG. 2A, the user developer is selecting Safari® on iPhone® 12. Referencing FIGS. 1 and 2A, once the user developer selects a type of a remote device **112** and a browser type RDB **109**, a physical remote device **112**, matching the user's selection is chosen in a datacenter **116**. The chosen remote device **112** is configured with the user selection of the RDB **109** and connected to the REL application **106** through the repeater **110** and the tunneling connection **108**.

(24) The software **103** can include any software application or program, including a web application, a website, or other software that a developer may wish to develop and test across multiple platforms. As an example, the software **103** can be a web application or a website,

executable and/or accessible on the RDB **109**. The user can enter a request in the form of a uniform resource locator (URL) pointing to a host server, hosting the software **103**. The user enters the URL in the replica display generated on the ULB **105**. The infrastructure **100** captures the entered URL in the replica display and injects it in an address bar field of the RDB **109**, causing the RDB **109** to issue a request to access the URL entered by the user in the replica display.

(25) FIG. **2B** illustrates a diagram of the ULB **105** and the RDB **109** during a test session of the software **103**. The software **103** in this example is a localhost website, accessible via a localhost server resident on the user's local machine. The user's local machine also executes the ULB **105**. The user can access the REL application **106** and the infrastructure **100** via entering a URL of the REL application **106** in the address bar **204** of the ULB **105**. However, in other implementations the REL application **106** can be a desktop application. The user developer can access various tools of the infrastructure **100** via a developer toolbar **208**. The developer toolbar **208** can include items to allow the user developer to enable or disable various features of the infrastructure **100**, relating to a test session. For example, enabling or disabling media injection features, enabling or disabling accessibility features, turning ON or OFF viewing of various debugging, session reporting and logging features of the infrastructure **100**.

(26) A localhost website is shown as an example for the software **103**, the software **103** can be any website accessible on the user's private test network **102** and/or accessible on a public network, such as the Internet. In some embodiments, the user of the infrastructure **100** can test accessing various URLs from the RDB **109** to test how those servers hosting the URLs operate on the RDB **109** and the remote device **112**. The user enters a URL request in the replica display **202** in the portion **206** of the replica display **202**, which corresponds to the address bar **210** in the RDB **109**. The infrastructure **100** captures the user's URL request entered into the portion **206**, via the REL application **106** and transmits the URL request to the remote device **112**. The URL request is injected into the address bar **210** in the RDB **109**. Consequently, the RDB **109** issues the URL request from the remote device **112**.

(27) Referencing FIG. **1**, the infrastructure **100** includes components to route traffic requests related to the user accessing various URLs from the RDB **109**. For example, in some implementations of the infrastructure **100**, a proxy server **115** can be installed on the host **118** to receive traffic requests, or URL access requests from the RDB **109**. The remote device **112** can be set up with proxy settings **130**, which directs the RDB **109** to proxy its traffic to the IP address and port of the host **118**. The proxy server **115** receives and forwards the requests to a repeater **110**. In some implementations, the repeater **110** is also a type of proxy server. Proxy servers act as intermediaries between request and response, and can provide a variety of functionality, including providing privacy, security, filtering, firewall, speed efficiency, caching, and other functionality. Proxy servers differ in how much exposure they have into the substance of the data they are routing. For example, some proxies implementing SSL inspection have a complete view into the content of the traffic they route, including passwords, and other sensitive data. Other proxy servers, for example, some using the TCP/IP model, may have no access to the payload in the data packets being transmitted, other than some routing information. In a preferred embodiment, both the proxy server **115** and the repeater **110** do not have access to payload or substantive data being transmitted, and only operate on the addresses contained in the data they transmit.

(28) The URLs can broadly include at least two categories. The first category of the user URL requests includes public URLs resolvable to a public host on a public network, such as the Internet. The second category of the user URL requests can include private URLs only resolvable on the user's private test network **102** and to a private host server in the user test network **102**. The repeater **110** receives the URL requests from the RDB **109** and resolves the requests. The repeater **110** forwards the public URL access requests to public servers **114**. The repeater **110** is not in the user's private test network **102** and cannot resolve private URL access requests. The repeater **110** forwards the private URL access requests to the user's private test network **102**. In some

embodiments, the REL application **106** receives the private URL access requests from the repeater **110**, and since the REL application **106** is running inside the user's private test network **102**, it can forward or otherwise resolve the private URL access requests within the user's private test network **102**. For example, the REL application **106** can resolve the private URL access requests to one or more user servers **104** inside the user's private test network **102**.

(29) Some browsers include an exclusion list of host URLs for security reasons. These browsers do not forward traffic requests for accessing host URLs on their exclusion list. For example, some browsers do not forward requests to access localhost servers to a proxy. In computer networking, a localhost is a host name that refers to the computer that is executing a program. The term is used when making a loopback request to the same computer. In other words, the localhost is a server running on a machine that is only accessible from that machine and refers back to the machine. Software developers use localhost and localhost servers to test and develop their programs. The IP address of a localhost is "127.0.0.1". In other words, typing localhost in the address bar of a browser sends a request from the browser to the "127.0.0.1" IP address. In the infrastructure **100**, some user's selection of RDB **109** do not forward URL requests to access a localhost to the proxy server **115**. A browser exclusion list can be alternatively referred to as the implicit bypass rules of the browser. When the browser's implicit bypass rules might otherwise prevent a user developer to issue a traffic request from the RDB **109**, the infrastructure **100** can modify the URL of the traffic request to avoid the exclusion list or implicit bypass rules of the browser.

(30) FIG. 3 illustrates a diagram of an example of operations and components of the infrastructure **100** to avoid an exclusion list of a browser running on the remote device **112**. The user is a developer of software utilizing the infrastructure **100** to conduct test and development activity. In this example, the user has configured a localhost **301** on the user's local machine **305** for the purpose of conducting test and development activity. The user's local machine **305** is part of the user's private test network **102**. Besides the localhost **301** running on the user's local machine **305**, the user's private test network **102** can also include one or more user servers **104** for the purpose of conducting test and development activity.

(31) The user can access the infrastructure **100**, by running the REL application **106** on the user's local machine **305**. When the REL application **106** is a browser-based web application, the user can run the REL application **106** by entering a URL provided by the operator of the infrastructure **100** in the address bar **204** of the ULB **105**. Using a developer toolbar **208**, the user can select a type of platform and a browser to conduct test and development. The infrastructure **100** can select a remote device **112** and an RDB **109**, matching the user's selection of the platform and the browser, respectively. The infrastructure **100** can generate a replica display **202** on ULB **105** by streaming a video feed of the remote device **112** to the ULB **105**. The user can enter URL requests in the portion **206** of the replica display **202**, and the infrastructure **100** can capture and inject the URL requests in the address bar **210** on RDB **109**. The RDB **109** can issue the URL requests by forwarding them to a proxy server **115**. The proxy server **115** can be resident on the host **118** and in communication with a repeater **110**. Proxy settings **130** on the remote device **112** can configure the remote device **112** and/or the RDB **109** to send traffic to the proxy server **115**. The repeater **110** determines whether the URL request is for a public server **114** and if yes, forwards the URL request to the public servers **114**, receives a response from the public servers **114** and forwards the response back to the remote device **112** and the RDB **109** through the proxy **115**. If the repeater **110** cannot resolve a URL request to a public server **114**, it assumes the URL request is directed to the user's private test network **102**, and forwards the private URL request to the test network **102**. The REL application **106** can receive the private URL request and resolve it to a user server **104** within the test network **102**. The REL application **106** can also forward the response from the user's server **104** back to the repeater, proxy, the remote device **112** and ultimately the RDB **109** through its tunneling connection.

(32) The RDB **109** can include implicit bypass rules in the form of an exclusion list **302**. The URL

requests matching the URLs on the exclusion list **302** can be restricted based on one or more rules. For example, the RDB **109** may block forwarding the local host access requests to the proxy **115**. In this case, the exclusion list **302** can include URL requests, such as “localhost,” “*.localhost,” “loopback,” “[: 1],” and other similar URLs. If a user's request injected in the address bar **210** includes one of these items, the RDB **109** does not forward that user request to the proxy **115**. Other browsers that do not have these items on their exclusion list or lack implicit bypass rules forward requests to access localhost to proxy **115** and do not block such traffic.

(33) In some embodiments, the RDB **109** can be modified or configured with an auxiliary code or a browser extension **304** to avoid triggering the implicit bypass rules of the RDB **109**. The browser extension **304** can modify a URL request injected in the address bar **210** to avoid triggering the bypass rules of the RDB **109**. For example, for URL requests matching the exclusion list **302**, the browser extension **304** can replace the URL requests with an alias that does not match the URLs on the exclusion list, but points to the same IP address of the original URL. In this manner, the user does not have to modify his workflow and can continue to enter the same URL, even one that is on an exclusion list of a browser, and the browser extension **304** can modify the URL request to avoid triggering the exclusion list of the browser. In some embodiments, the browser extension **304** can include a browser event monitor **303**, which listens for browser events and when a new URL request is entered, it can trigger the browser extension **304** to parse the request to detect if any host server name on the exclusion list **302** is embedded in the request. Upon detecting an excluded host server name, the browser extension **304** can replace the host name with an alias URL.

(34) The operator of the infrastructure **100** can provide one or more alias URLs with host names that do not match any items on the exclusion list **302**, but nevertheless point to the same IP address of an item on the exclusion list **302**. For example, the URL request “localhost,” which may be on the exclusion list **302** points to the IP address “127.0.0.1.” The operator of the infrastructure **100** can provide an alias URL, for example “bs-local.com,” which is not on the exclusion list **302**, but points to the same IP address “127.0.0.1.” When the RDB **109** receives the URL request “bs-local.com,” the implicit bypass rules are not triggered and the RDB **109** forwards the request to the proxy server **115** and the remaining operations of the infrastructure **100** proceed as they would for any other URL requests. In this example, the repeater **110** cannot resolve the URL request “bs-local.com,” and forwards it to the REL application **106**. The REL application **106** forwards the request to the localhost **301** because the IP address of the domain name or host server “bs-local.com” points to the IP address of the localhost **301**.

(35) Some URL requests injected into the address bar **210** include a host name and a port number. For example, the localhost **301**, can be accessed via a URL request “localhost:8000” from the user's local machine **305**. The term “localhost” is the host server name, and the number 8000 is the port or the communication endpoint to which this example URL request is pointed. As described earlier, the host server name or URL localhost can be on the exclusion list **302**. The browser extension **304** replaces the host server name “localhost,” which is on the exclusion list **302** with an alias URL “bs-local.com,” but leaves the port information intact. In this manner, the request reaches the same communication endpoint as intended by the user.

(36) In other words, in some cases, a URL request **306** includes localhost URL (for example, “localhost”) and a port number. The browser extension **304** parses the URL request **306** and generates a modified URL request **308**. The modified URL request **308** includes an alias URL replacing the localhost URL and the same port number as in the original URL request **306**. For example, the original URL request **306** entered by the user can be “localhost:8000,” and the modified URL request **308** is “bs-local.com:8000.” The URL request can include other portions in addition to or instead of a port number. For example, some URL requests include a localhost URL plus path parameters. In these and similar scenarios, the browser extension **304** generates the modified URL request **308** in the same manner, as described above. To generate the modified URL request, the browser extension **304** replaces the localhost portion of the URL request with an alias

URL and concatenates and/or adds the additional elements or portions of the original URL request to the alias URL unchanged. For example, a URL request “localhost:8000/health/” can be modified to “bs-local.com:8000/health/.”

(37) Furthermore, in some embodiments, the browser extension **304** operates by detecting that a URL request has been restricted by the RDB **109** as a result of triggering the bypass rules of the RDB **109**. The browser extension **304** generates the modified URL request as described above, by replacing the restricted URL portion with an alias URL, not on the exclusion list **302**, and forwards the modified URL request to the proxy server **115**. In other words, in some implementations, an initial URL request **306** is issued by the RDB **109** and subsequently restricted. The browser extension **304** detects the restriction and generates a modified URL request **308** and forwards the modified URL request **308** to the proxy server **115**.

(38) FIG. 4 illustrates an example method **400** of operation of the infrastructure **100**. The method starts at step **402**. At step **404**, a communication network between a user local machine **305** and a remote device **112** is established. In one implementation, the communication network can be a peer-to-peer (P2P) communication network, but other communication network methodologies are also possible. The communication network can include a data channel and a video channel. At step **406**, the video channel is used to stream a video feed of the display of the remote device **112** to the user local machine **305**. At step **408**, the infrastructure **100** can capture user interactions with the replica display and transmit them via the data channel to the remote device **112**. In some implementations, the user interactions can be captured by the ULB **105**. In some embodiments, the user interactions can be entered into a browser running on the remote device. For example, the user interactions can include the user entering URL requests to a portion **206** of the replica display **202** corresponding to an address bar **210** of the RDB **109**. At step **410**, the browser running on the remote device **112** can issue traffic requests, for example, URL access requests.

(39) At step **412**, the infrastructure **100** can route the traffic requests issued from the browser running on the remote device **112**. For example, the requests can be received by a proxy server **115** and routed to a repeater **110**. The repeater **110** resolves the public server request and routes the remaining requests to a private test network **102** of the user. The responses from the public servers or the private servers of the user network are routed through the repeater and the proxy server back to the remote device and the browser running on the remote device. The method ends at step **414**.

(40) FIG. 5 illustrates an example method **500** of avoiding the implicit bypass rules of a browser. The method starts at step **502**. The browser running on the remote device **112** can include implicit bypass rules, which restricts forwarding or proxying URL requests for certain host servers or domains on an exclusion or restriction list. For example, the browser running on the remote device **112** can restrict proxying URL requests to a localhost. At step **504**, a browser extension can detect whether a URL request entered into the browser running on the remote device **112** includes a request for access to a restricted URL found on the browser's exclusion or restriction list. The browser extension can parse the URL requests entered into the browser running on the remote device **112** and match them against the exclusion or restriction list to make that determination. In another embodiment, the URL request can trigger the bypass rules or exclusion list of the browser, and the browser extension can detect the exclusionary action of the browser. Upon detecting the exclusionary action of the browser, the browser extension can receive and modify the URL request.

(41) At step **506**, upon determining that the URL requests includes a restricted URL access request, the browser extension can modify the URL request and replace the restricted URL with an alias URL, which has the same IP address of the restricted URL. In some embodiments, the URL request includes a restricted URL portion and a port number. The browser extension can replace the restricted URL portion with an alias URL and leave the port number unchanged. In some embodiments, the restricted URL can include other or additional portions than the port number. For example, a restricted URL access can include portions such as path parameters or other portions depending on the application. In such a scenario, the modification of the URL request can include

replacing the restricted URL with the alias URL and concatenating and/or adding the remaining portions into the alias URL to construct the modified URL. For example, “localhost:8000/health/” can be modified to bs-local.com:8000/health/.” The browser extension enters the modified URL to the browser, and the browser routes the modified URL. In another embodiment, where the browser extension catches the restricted URL request after an initial exclusionary action of the browser upon the original request, the browser can send the modified URL. In other words, the browser extension can send the modified request after an exclusionary action upon the original URL request. The modified URL has the same IP address and communication endpoint (e.g., port number) as the original URL request and will reach the same communication endpoint as intended in the user request in the original URL request.

(42) An example of a restricted URL is a localhost server hosted and exposed only via a port number on the user's local machine. Some browsers running on the remote device **112** do not forward requests for access to a localhost to a proxy server. In other words, some browsers do not proxy a URL request to access a localhost. However, the same browsers can route the modified URL, as the modified URL is not on a restriction or exclusion list of the browser. In the case of a proxying restriction of a localhost, the browser extension can replace the host server, “localhost” with an alias URL (e.g., “bs-local.com”) that has the same IP address as the localhost (e.g., 127.0.0.1) and leave any port number unchanged. For example, the URL request “localhost:54455” can be modified to “bs-local.com:54455). At step **508**, the infrastructure **100** routes the modified URL, which reaches the same communication endpoint as intended in the user's original URL request. The method ends at step **510**.

(43) FIG. **6** illustrates an example method **600** of routing traffic and URL requests in the infrastructure **100**. The method starts at step **602**. At step **604**, the browser running on the remote device **112** forwards a URL request to a proxy server. The browser running on the remote device **112** receives the URL request from a user interaction entered on a replica display of the remote device **112** on the user's local machine **305**. Before forwarding the request, a browser extension can detect whether the URL request is for a URL on an exclusion or restriction list of the browser and can modify the URL request with an alias URL if the original URL request includes a host server or URL on the exclusion or restriction list. The alias URL has the same IP address and communication endpoint as the restricted URL. If the URL request is modified, the browser forwards the modified URL request. At step **606**, the proxy server forwards the URL request to a repeater. At step **608**, the repeater routes the request based on whether the request is directed to a public server. For example, the repeater forwards the public host server requests to a public host server, and if the repeater cannot resolve a URL request to a public server, it forwards the URL request to the user private test network **102**. At step **610**, a REL application **106** can receive the non-public server requests and resolve it based on domain name system (DNS) rules of the user's private test network **102**. In the example of localhost, the REL application can resolve the modified URL requests pointing to a localhost on the same machine as the REL application's machine. The method ends at step **612**.

(44) Example Implementation Mechanism Hardware Overview

(45) Some embodiments are implemented by a computer system or a network of computer systems. A computer system may include a processor, a memory, and a non-transitory computer-readable medium. The memory and non-transitory medium may store instructions for performing methods, steps and techniques described herein.

(46) According to one embodiment, the techniques described herein are implemented by one or more special-purpose computing devices. The special-purpose computing devices may be hard-wired to perform the techniques, or may include digital electronic devices such as one or more application-specific integrated circuits (ASICs) or field programmable gate arrays (FPGAs) that are persistently programmed to perform the techniques, or may include one or more general purpose hardware processors programmed to perform the techniques pursuant to program instructions in firmware, memory, other storage, or a combination. Such special-purpose computing devices may

also combine custom hard-wired logic, ASICs, or FPGAs with custom programming to accomplish the techniques. The special-purpose computing devices may be server computers, cloud computing computers, desktop computer systems, portable computer systems, handheld devices, networking devices or any other device that incorporates hard-wired and/or program logic to implement the techniques.

(47) For example, FIG. 7 is a block diagram that illustrates a computer system **1000** upon which an embodiment of can be implemented. Computer system **1000** includes a bus **1002** or other communication mechanism for communicating information, and a hardware processor **1004** coupled with bus **1002** for processing information. Hardware processor **1004** may be, for example, special-purpose microprocessor optimized for handling audio and video streams generated, transmitted or received in video conferencing architectures.

(48) Computer system **1000** also includes a main memory **1006**, such as a random access memory (RAM) or other dynamic storage device, coupled to bus **1002** for storing information and instructions to be executed by processor **1004**. Main memory **1006** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **1004**. Such instructions, when stored in non-transitory storage media accessible to processor **1004**, render computer system **1000** into a special-purpose machine that is customized to perform the operations specified in the instructions.

(49) Computer system **1000** further includes a read only memory (ROM) **1008** or other static storage device coupled to bus **1002** for storing static information and instructions for processor **1004**. A storage device **1010**, such as a magnetic disk, optical disk, or solid state disk is provided and coupled to bus **1002** for storing information and instructions.

(50) Computer system **1000** may be coupled via bus **1002** to a display **1012**, such as a cathode ray tube (CRT), liquid crystal display (LCD), organic light-emitting diode (OLED), or a touchscreen for displaying information to a computer user. An input device **1014**, including alphanumeric and other keys (e.g., in a touch screen display) is coupled to bus **1002** for communicating information and command selections to processor **1004**. Another type of user input device is cursor control **1016**, such as a mouse, a trackball, or cursor direction keys for communicating direction information and command selections to processor **1004** and for controlling cursor movement on display **1012**. This input device typically has two degrees of freedom in two axes, a first axis (e.g., x) and a second axis (e.g., y), that allows the device to specify positions in a plane. In some embodiments, the user input device **1014** and/or the cursor control **1016** can be implemented in the display **1012** for example, via a touch-screen interface that serves as both output display and input device.

(51) Computer system **1000** may implement the techniques described herein using customized hard-wired logic, one or more ASICs or FPGAs, firmware and/or program logic which in combination with the computer system causes or programs computer system **1000** to be a special-purpose machine. According to one embodiment, the techniques herein are performed by computer system **1000** in response to processor **1004** executing one or more sequences of one or more instructions contained in main memory **1006**. Such instructions may be read into main memory **1006** from another storage medium, such as storage device **1010**. Execution of the sequences of instructions contained in main memory **1006** causes processor **1004** to perform the process steps described herein. In alternative embodiments, hard-wired circuitry may be used in place of or in combination with software instructions.

(52) The term “storage media” as used herein refers to any non-transitory media that store data and/or instructions that cause a machine to operation in a specific fashion. Such storage media may comprise non-volatile media and/or volatile media. Non-volatile media includes, for example, optical, magnetic, and/or solid-state disks, such as storage device **1010**. Volatile media includes dynamic memory, such as main memory **1006**. Common forms of storage media include, for example, a floppy disk, a flexible disk, hard disk, solid state drive, magnetic tape, or any other

magnetic data storage medium, a CD-ROM, any other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, and EPROM, a FLASH-EPROM, NVRAM, any other memory chip or cartridge.

(53) Storage media is distinct from but may be used in conjunction with transmission media. Transmission media participates in transferring information between storage media. For example, transmission media includes coaxial cables, copper wire and fiber optics, including the wires that comprise bus **1002**. Transmission media can also take the form of acoustic or light waves, such as those generated during radio-wave and infra-red data communications.

(54) Various forms of media may be involved in carrying one or more sequences of one or more instructions to processor **1004** for execution. For example, the instructions may initially be carried on a magnetic disk or solid state drive of a remote computer. The remote computer can load the instructions into its dynamic memory and send the instructions over a telephone line using a modem. A modem local to computer system **1000** can receive the data on the telephone line and use an infra-red transmitter to convert the data to an infra-red signal. An infra-red detector can receive the data carried in the infra-red signal and appropriate circuitry can place the data on bus **1002**. Bus **1002** carries the data to main memory **1006**, from which processor **1004** retrieves and executes the instructions. The instructions received by main memory **1006** may optionally be stored on storage device **1010** either before or after execution by processor **1004**.

(55) Computer system **1000** also includes a communication interface **1018** coupled to bus **1002**. Communication interface **1018** provides a two-way data communication coupling to a network link **1020** that is connected to a local network **1022**. For example, communication interface **1018** may be an integrated services digital network (ISDN) card, cable modem, satellite modem, or a modem to provide a data communication connection to a corresponding type of telephone line. As another example, communication interface **1018** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN. Wireless links may also be implemented. In any such implementation, communication interface **1018** sends and receives electrical, electromagnetic or optical signals that carry digital data streams representing various types of information.

(56) Network link **1020** typically provides data communication through one or more networks to other data devices. For example, network link **1020** may provide a connection through local network **1022** to a host computer **1024** or to data equipment operated by an Internet Service Provider (ISP) **1026**. ISP **1026** in turn provides data communication services through the world wide packet data communication network now commonly referred to as the “Internet” **1028**. Local network **1022** and Internet **1028** both use electrical, electromagnetic or optical signals that carry digital data streams. The signals through the various networks and the signals on network link **1020** and through communication interface **1018**, which carry the digital data to and from computer system **1000**, are example forms of transmission media.

(57) Computer system **1000** can send messages and receive data, including program code, through the network(s), network link **1020** and communication interface **1018**. In the Internet example, a server **1030** might transmit a requested code for an application program through Internet **1028**, ISP **1026**, local network **1022** and communication interface **1018**.

(58) The received code may be executed by processor **1004** as it is received, and/or stored in storage device **1010**, or other non-volatile storage for later execution.

EXAMPLES

(59) It will be appreciated that the present disclosure may include any one and up to all of the following examples.

(60) Example 1: A method comprising: establishing a communication network between a first machine and a second machine; executing a browser on the second machine; generating a replica display of the second machine on the first machine; streaming a video feed of a display of the second machine to the first machine; displaying the video feed in the replica display; capturing user interactions with the replica display, wherein the user interactions comprise requests to access host

servers; entering the user interactions in an address bar of the browser; issuing the requests from the browser; and routing the requests to the host servers.

(61) Example 2: The method of Example 2, wherein the requests comprise a request to access a host URL on an exclusion list of the browser, and the browser is further configured to replace the request with an alias URL pointing to an IP address of the host URL on the exclusion list.

(62) Example 3: The method of some or all of Examples 1 and 2, wherein the requests comprise a request to access a localhost server on an exclusion list of the browser, and the browser is configured to replace the request with an alias URL pointing to the IP address of the localhost.

(63) Example 4: The method of some or all of Examples 1-3, wherein the first machine is connected to a private communication network and comprises a localhost server wherein the requests comprise a URL of the localhost server portion, wherein the localhost server is on an exclusion list of the browser running on the second machine, and the browser is configured to replace the URL of the localhost server portion of the request with an alias URL pointing to an IP address of the localhost server.

(64) Example 5: The method of some or all of Examples 1-4, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server, wherein the browser comprises an exclusion list of host URLs, wherein the browser does not forward requests for accessing host URLs on the exclusion list, to the proxy server; configuring the browser with an extension, wherein the extension is configured to: detect a request to access a host URL on the exclusion list; and replace the request to access a host URL on the exclusion list with an alias URL, not on the exclusion list, wherein the alias URL points to an IP address of a host URL on the exclusion list.

(65) Example 6: The method of some or all of Examples 1-5, wherein the first machine is connected to a private communication network comprising private host servers, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server; the proxy server forwarding the requests to a repeater; the repeater determining whether a request comprises a request to access a public host server, and routing the request to access the public host server to the public host server; and the repeater determining whether a request is not directed to access a public host server, and routing the request to the private communication network, wherein the private communication network routes the request to a private host server, wherein the first machine runs a localhost server and the browser comprises an exclusion list of host URLs, wherein the browser does not forward to the proxy server, requests comprising the host URLs on the exclusion list, and the method further comprises: configuring a browser extension to receive a request for access to a URL on the exclusion list; modifying the request with an alias URL, the alias URL pointing to an IP address of the localhost server; and the browser forwarding the modified request to the proxy server.

(66) Example 7: The method of some or all of Example 1-6, further comprising: detecting an exclusionary action of the browser comprising the browser restricting forwarding of a request to a host server on an exclusion list of the browser; modifying the request; and routing the modified request to the host server.

(67) Example 8: A non-transitory computer storage that stores executable program instructions that, when executed by one or more computing devices, configure the one or more computing devices to perform operations comprising: establishing a communication network between a first machine and a second machine; executing a browser on the second machine; generating a replica display of the second machine on the first machine; streaming a video feed of a display of the second machine to the first machine; displaying the video feed in the replica display; capturing user interactions with the replica display, wherein the user interactions comprise requests to access host servers; entering the user interactions in an address bar of the browser; issuing the requests from the browser; and routing the requests to the host servers.

(68) Example 9: The non-transitory computer storage of Example 9, wherein the requests comprise a request to access a host URL on an exclusion list of the browser, and the browser is further

configured to replace the request with an alias URL pointing to an IP address of the host URL on the exclusion list.

(69) Example 10: The non-transitory computer storage of some or all of Examples 8 and 9, wherein the requests comprise a request to access a localhost server on an exclusion list of the browser, and the browser is configured to replace the request with an alias URL pointing to the IP address of the localhost.

(70) Example 11: The non-transitory computer storage of some or all of Examples 8-10, wherein the first machine is connected to a private communication network and comprises a localhost server wherein the requests comprise a URL of the localhost server portion, wherein the localhost server is on an exclusion list of the browser running on the second machine, and the browser is configured to replace the URL of the localhost server portion of the request with an alias URL pointing to an IP address of the localhost server.

(71) Example 12: The non-transitory computer storage of some or all of Examples 8-11, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server, wherein the browser comprises an exclusion list of host URLs, wherein the browser does not forward requests for accessing host URLs on the exclusion list, to the proxy server, wherein the operations further comprise configuring the browser with an extension, wherein the extension is configured to: detect a request to access a host URL on the exclusion list; and replace the request to access a host URL on the exclusion list with an alias URL, not on the exclusion list, wherein the alias URL points to an IP address of a host URL on the exclusion list.

(72) Example 13: The non-transitory computer storage of some or all of Examples 8-12, wherein the first machine is connected to a private communication network comprising private host servers, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server; the proxy server forwarding the requests to a repeater; the repeater determining whether a request comprises a request to access a public host server, and routing the request to access the public host server to the public host server; and the repeater determining whether a request is not directed to access a public host server, and routing the request to the private communication network, wherein the private communication network routes the request to a private host server, wherein the first machine runs a localhost server and the browser comprises an exclusion list of host URLs, wherein the browser does not forward to the proxy server, requests comprising the host URLs on the exclusion list, and the operations further comprise: configuring a browser extension to receive a request for access to a URL on the exclusion list; modifying the request with an alias URL, the alias URL pointing to an IP address of the localhost server; and the browser forwarding the modified request to the proxy server.

(73) Example 14: The non-transitory computer storage of some or all of Examples 8-13, wherein the operations further comprise: detecting an exclusionary action of the browser comprising the browser restricting forwarding of a request to a host server on an exclusion list of the browser; modifying the request; and routing the modified request to the host server.

(74) Example 15: A system comprising a processor, the processor configured to perform operations comprising: establishing a communication network between a first machine and a second machine; executing a browser on the second machine; generating a replica display of the second machine on the first machine; streaming a video feed of a display of the second machine to the first machine; displaying the video feed in the replica display; capturing user interactions with the replica display, wherein the user interactions comprise requests to access host servers; entering the user interactions in an address bar of the browser; issuing the requests from the browser; and routing the requests to the host servers.

(75) Example 16: The system of Example 15, wherein the requests comprise a request to access a host URL on an exclusion list of the browser, and the browser is further configured to replace the request with an alias URL pointing to an IP address of the host URL on the exclusion list.

(76) Example 17: The system of some or all of Examples 15 and 16, wherein the requests comprise

a request to access a localhost server on an exclusion list of the browser, and the browser is configured to replace the request with an alias URL pointing to the IP address of the localhost. (77) Example 18: The system of some or all of Examples 15-17, wherein the first machine is connected to a private communication network and comprises a localhost server wherein the requests comprise a URL of the localhost server portion, wherein the localhost server is on an exclusion list of the browser running on the second machine, and the browser is configured to replace the URL of the localhost server portion of the request with an alias URL pointing to an IP address of the localhost server.

(78) Example 19: The system of some or all of Examples 15-18, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server, wherein the browser comprises an exclusion list of host URLs, wherein the browser does not forward requests for accessing host URLs on the exclusion list, to the proxy server, wherein the operations further comprise configuring the browser with an extension, wherein the extension is configured to: detect a request to access a host URL on the exclusion list; and replace the request to access a host URL on the exclusion list with an alias URL, not on the exclusion list, wherein the alias URL points to an IP address of a host URL on the exclusion list.

(79) Example 20: The system of some or all of Examples 15-19, wherein the first machine is connected to a private communication network comprising private host servers, wherein routing the requests further comprises: the browser forwarding the requests to a proxy server; the proxy server forwarding the requests to a repeater; the repeater determining whether a request comprises a request to access a public host server, and routing the request to access the public host server to the public host server; and the repeater determining whether a request is not directed to access a public host server, and routing the request to the private communication network, wherein the private communication network routes the request to a private host server, wherein the first machine runs a localhost server and the browser comprises an exclusion list of host URLs, wherein the browser does not forward to the proxy server, requests comprising the host URLs on the exclusion list, and the operations further comprise: configuring a browser extension to receive a request for access to a URL on the exclusion list; modifying the request with an alias URL, the alias URL pointing to an IP address of the localhost server; and the browser forwarding the modified request to the proxy server.

(80) While the invention has been particularly shown and described with reference to specific embodiments thereof, it should be understood that changes in the form and details of the disclosed embodiments may be made without departing from the scope of the invention. Although various advantages, aspects, and objects of the present invention have been discussed herein with reference to various embodiments, it will be understood that the scope of the invention should not be limited by reference to such advantages, aspects, and objects. Rather, the scope of the invention should be determined with reference to patent claims.

Claims

1. A method comprising: establishing a communication network between a local machine and a host machine coupled to a remote device, the remote device running a browser, wherein the browser comprises a URL exclusion list, wherein the remote device comprises proxy settings and is configured to direct traffic to a proxy server comprised in the host machine, wherein the browser does not forward URL access requests to the proxy server when the URL access requests are on the URL exclusion list; capturing user interactions on a replica display, the replica display, generated on the local machine, from a video feed of a display of the remote device, wherein the user interactions comprise a URL access request to access a host server, wherein the URL access request corresponds to an IP address of the host server; detecting when the URL access request is on the URL exclusion list; in response to detecting the URL access request on the URL exclusion list,

modifying, with a browser extension of the browser, the URL access request by replacing the URL in the URL access request with an alias URL, wherein the alias URL corresponds to the IP address of the host server; issuing the modified URL access request from the browser; and routing the modified URL access request from the remote device to the IP address of the host server.

2. The method of claim 1, wherein the local machine is at a first location, and the host machine and the remote device are in a datacenter at a second location, the second location being remote relative to the first location.
3. The method of claim 1, wherein the URL access request comprises a request to access a localhost server on the URL exclusion list, and the browser is configured to replace the request with an alias URL corresponding to an IP address of the localhost.
4. The method of claim 1, wherein the local machine is coupled to a private network and comprises a localhost server wherein the URL access request comprises a URL of the localhost server portion, wherein the URL localhost server portion is on the URL exclusion list, and the browser is configured to replace the URL of the localhost server portion of the URL access request with an alias URL portion corresponding to an IP address of the localhost server.
5. The method of claim 4, wherein the URL access request further comprises a communication path portion, and the browser is configured to leave the communication path portion unchanged.
6. The method of claim 1, wherein the method further comprises: detecting the browser not forwarding the URL access request to the proxy server.
7. The method of claim 1, wherein the proxy server is configured to route traffic from the remote device, wherein the method further comprises: configuring the browser, with the browser extension, to detect when the URL access request is on the URL exclusion list; modifying the URL access request, by replacing the URL in the URL access request with an alias URL, wherein the alias URL corresponds to the IP address of the host server, wherein routing the modified URL access request further comprises: forwarding the modified URL access request to the proxy server.
8. The method of claim 7, wherein the local machine is coupled to a private network, the private network comprising a private server, configured to resolve URL access requests, wherein the local machine comprises a localhost server, wherein routing the modified URL access request further comprises: the proxy server forwarding the modified URL access request to a repeater; the repeater determining whether the modified URL access request comprises a request to access a public server, and routing the request to access the public server to the public server; and the repeater determining whether the modified URL access request is not directed to access a public server, and routing the request to the private network, wherein the private network routes the request to the private server, wherein the private server forwards the modified URL access request to the localhost.
9. A non-transitory computer storage medium that stores executable program instructions that, when executed by one or more computing devices, configure the one or more computing devices to perform operations comprising: establishing a communication network between a local machine and a host machine coupled to a remote device, the remote device running a browser, wherein the browser comprises a URL exclusion list, wherein the remote device comprises proxy settings and is configured to direct traffic to a proxy server comprised in the host machine, wherein the browser does not forward URL access requests to the proxy server when the URL access requests are on the URL exclusion list; capturing user interactions on a replica display, the replica display, generated on the local machine, from a video feed of a display of the remote device, wherein the user interactions comprise a URL access request to access a host server, wherein the URL access request corresponds to an IP address of the host server; detecting when the URL access request is on the URL exclusion list; in response to detecting the URL access request on the URL exclusion list, modifying, with a browser extension of the browser, the URL access request by replacing the URL in the URL access request with an alias URL, wherein the alias URL corresponds to the IP address of the host server; issuing the modified URL access request from the browser; and routing the

modified URL access request from the remote device to the IP address of the host server.

10. The non-transitory computer storage medium of claim 9, wherein the local machine is at a first location, and the host machine and the remote device are in a datacenter at a second location, the second location being remote relative to the first location.

11. The non-transitory computer storage medium of claim 9, wherein the URL access request comprises a request to access a localhost server on the URL exclusion list, and the browser is configured to replace the request with an alias URL corresponding to an IP address of the localhost.

12. The non-transitory computer storage medium of claim 9, wherein the local machine is coupled to a private network and comprises a localhost server wherein the URL access request comprises a URL of the localhost server portion, wherein the URL localhost server portion is on the URL exclusion list, and the browser is configured to replace the URL of the localhost server portion of the URL access request with an alias URL portion corresponding to an IP address of the localhost server.

13. The non-transitory computer storage medium of claim 12, wherein the URL access request further comprises a communication path portion, and the browser is configured to leave the communication path portion unchanged.

14. The non-transitory computer storage medium of claim 9, wherein the operations further comprise: detecting the browser not forwarding the URL access request to the proxy server.

15. The non-transitory computer storage medium of claim 9, wherein the proxy server is configured to route traffic from the remote device, wherein the operations further comprise: configuring the browser, with the browser extension, to detect when the URL access request is on the URL exclusion list; modifying the URL access request, by replacing the URL in the URL access request with an alias URL, wherein the alias URL corresponds to the IP address of the host server, wherein routing the modified URL access request further comprises: forwarding the modified URL access request to the proxy server.

16. The non-transitory computer storage medium of claim 15, wherein the local machine is coupled to a private network, the private network comprising a private server, configured to resolve URL access requests, wherein the local machine comprises a localhost server, wherein routing the modified URL access request further comprises: the proxy server forwarding the modified URL access request to a repeater; the repeater determining whether the modified URL access request comprises a request to access a public server, and routing the request to access the public server to the public server; and the repeater determining whether the modified URL access request is not directed to access a public server, and routing the request to the private network, wherein the private network routes the request to the private server, wherein the private server forwards the modified URL access request to the localhost.

17. A system comprising one or more processors, the one or more processors configured to perform operations comprising: establishing a communication network between a local machine and a host machine coupled to a remote device, the remote device running a browser, wherein the browser comprises a URL exclusion list, wherein the remote device comprises proxy settings and is configured to direct traffic to a proxy server comprised in the host machine, wherein the browser does not forward URL access requests to the proxy server when the URL access requests are on the URL exclusion list; capturing user interactions on a replica display, the replica display, generated on the local machine, from a video feed of a display of the remote device, wherein the user interactions comprise a URL access request to access a host server, wherein the URL access request corresponds to an IP address of the host server; detecting when the URL access request is on the URL exclusion list; in response to detecting the URL access request on the URL exclusion list, modifying, with a browser extension of the browser, the URL access request by replacing the URL in the URL access request with an alias URL, wherein the alias URL corresponds to the IP address of the host server; issuing the modified URL access request from the browser; and routing the modified URL access request from the remote device to the IP address of the host server.

18. The system of claim 17, wherein the URL access request comprises a request to access a localhost server on the URL exclusion list, and the browser is configured to replace the request with an alias URL corresponding to an IP address of the localhost.
